Materials and Design 196 (2020) 109098

Contents lists available at ScienceDirect

Materials and Design

journal homepage: www.elsevier.com/locate/matdes

Deep learning for topology optimization of 2D metamaterials

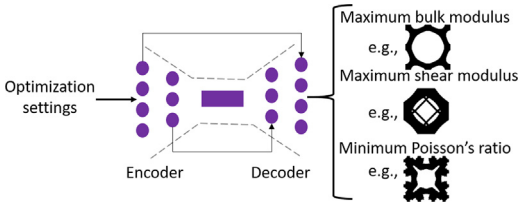
Hunter T. Kollmann [a], Diab W. Abueidda [a,b,*], Seid Koric [a,b,*], Erman Guleryuz [b], Nahil A. Sobh [c]

[a] Department of Mechanical Science and Engineering, University of Illinois at Urbana-Champaign, United States of America
[b] National Center for Supercomputing Applications, University of Illinois at Urbana-Champaign, United States of America
[c] Beckman Institute of Science and Technology, University of Illinois at Urbana-Champaign, United States of America

HIGHLIGHTS

- A deep learning (DL) model is developed for obtaining optimized metamaterials.
- The DL model optimizes for bulk modulus, shear modulus, or Poisson's ratio.
- Parallel computing on the HPC system is used to generate the data needed for the DL model training.
- The developed DL model shows high accuracy.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 15 July 2020
Received in revised form 13 August 2020
Accepted 24 August 2020
Available online 29 August 2020

Keywords:
Architected materials
Auxetic materials
Homogenization
Machine learning
Microstructure
Periodic boundary conditions (PBCs)

ABSTRACT

Data-driven models are rising as an auspicious method for the geometrical design of materials and structural systems. Nevertheless, existing data-driven models customarily address the optimization of structural designs rather than metamaterial designs. Metamaterials are emerging as promising materials exhibiting tailorable and unprecedented properties for a wide spectrum of applications. In this paper, we develop a deep learning (DL) model based on a convolutional neural network (CNN) that predicts optimal metamaterial designs. The developed DL model non-iteratively optimizes metamaterials for either maximizing the bulk modulus, maximizing the shear modulus, or minimizing the Poisson's ratio (including negative values). The data are generated by solving a large set of inverse homogenization boundary values problems, with randomly generated geometrical features from a specific distribution. Such s data-driven model can play a vital role in accelerating more computationally expensive design problems, such as multiscale metamaterial systems.

© 2020 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. Introduction

Mechanical metamaterials are materials with unique architectures leading to mechanical properties unattainable by their constituent material(s) [1]. Materials with enhanced properties have been intensely sought after in the scientific and industrial communities [2–8]. These materials are usually optimized through the choice of the constituent materials, the volume fraction, and the architecture. The effects of constituent materials and volume fraction have been extensively studied and are relatively mature, so there is limited progress that can be efficiently made utilizing these two factors [9]. Nevertheless, designing the architectures of the material is contemporary, as it efficiently allows obtaining materials with unprecedented properties not found in natural materials. These properties include and not limited to very high stiffness-to-weight ratio and negative Poisson's Ratio [10,11]. These materials with enhanced properties are usually attained through intuitions and experiments, bioinspiration, and/or topology optimization [12–14]. One application of topology optimization algorithms that is gaining a rising interest is the design of metamaterials. Topology optimization offers a systematic method to design materials given a set of loads,

* Corresponding authors at: National Center for Supercomputing Applications, University of Illinois at Urbana-Champaign, United States of America.
E-mail addresses: abueidd2@illinois.edu (D.W. Abueidda), koric@illinois.edu (S. Koric).

https://doi.org/10.1016/j.matdes.2020.109098
0264-1275/© 2020 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

boundary conditions, and constraints. Topology optimization aims to identify the optimal material distribution that provides the highest performance of the system while keeping the design constraints satisfied [15,16].

Until recently, many of the complex geometries were not feasible to make. Advances in additive manufacturing make the fabrication of these materials with complex architectures possible [17–21]. While more advanced fabrication techniques allow for new opportunities for making such metamaterials, the optimization problem is challenging [22]. More specifically, the topology optimization for metamaterial design is an inverse homogenization problem [23–25], which is a one-to-many mapping (from property to microstructures) and has an infinite-dimensional design space. Huang et al. [26] used the bidirectional evolutionary structural optimization (BESO) technique to maximize bulk or shear modulus. Materials with negative Poisson's ratios (auxetic materials) have a wide array of industrial and engineering applications. Auxetic materials can be used for crash energy absorption because they can become thicker (thinner) when stretched (compressed) so the energy is absorbed, and the material warps in a more desirable direction. They can also be used to increase fracture toughness, resistance to shear, and acoustic energy absorption [27–30]. Auxetic materials can also deform through the rotation of its ligaments, which are classified as chiral auxetic metamaterials [27]. Also, Vogiatzis et al. [31] proposed a level set based method for topology optimization of negative Poisson's ratio metamaterials with both single- and multiple-material(s). Mechanical metamaterials play a vital role in designing multiscale materials and structural systems [22,32].

Numerical analyses, including topology optimization problems, can be computationally expensive due to the many required iterations. Researchers have recently adopted many machine learning (ML) algorithms to almost instantly (without iterations) obtain solutions for different solid mechanics problems, even at low-end machines such as laptops. ML has been an effective computational tool in image and speech recognition, medical diagnoses, autopilot in automotive scenarios, and financial services [33–35]. Solid mechanics and materials science are no exception. With proper descriptors (e.g., geometry, material properties, and/or loading conditions), adequate training data, and appropriate ML architectures, ML models can be trained for obtaining the responses of materials and structural systems.

Several deep learning (DL) algorithms have been developed to study the mechanical properties of composites [36,37], the plastic response of different systems [38,39], the thermal conductivity of composites [40], the phononic properties [41], the effect of flexoelectricity on nanostructures [42], the defects of materials [43], the fatigue of materials [44], topology optimization [45], etc. Yang et al. [46] combined the principal component analysis (PCA) with convolutional neural networks (CNN) to predict the entire stress-strain curve of checkerboard composite failures beyond the elastic limit. Additionally, Nguyen et al. [47] utilized a neural network model to obtain the strength of foamed concrete. Data-driven models have shown broad applicability to accelerate design processes and materials discovery [36,42,48]. Also, physics-informed neural network models have been recently developed to study hyperelastic and other physical phenomena [49–51].

Both topology optimization types, gradient-free (e.g., genetic algorithms) and gradient-based, require a relatively large number of function evaluations, i.e., solving many finite element (FE) problems, leading to the need of powerful resources and time-consuming computations. Gradient-free topology optimization algorithms are easier to implement, as one can avoid the mathematical derivation and computations required to obtain the gradients [36,52,53]. However, gradient-free optimization algorithms usually need a relatively large number of function evaluations, especially if the design space is enormous, which is the case in most topology optimization problems. Gradient-based optimization algorithms usually have fast convergence rates compared to gradient-free optimizers. Nevertheless, the sensitivity analyses of the objective functions and constraints are challenging, especially with nonlinear constraints (e.g., stress constraints [54,55]) and/or objective functions. Objective functions and constraints, requiring nonlinear analysis such as plasticity [56] and viscoelasticity [57], are more complicated than the case of linear elasticity. Also, the nonconvexity of most topology optimization problems makes the selection of the starting point is a critical step, as the solution would be dependent on the starting point with the possibility of getting trapped in local minima.

Additionally, several researchers sought after developing deep learning models that predict the optimal designs almost instantly (without iterations) [58]. Lin et al. [59] developed a CNN model to accelerate the topology optimization of the conductive heat transfer. Also, Rawat and Shen [60,61] developed a generative adversarial network composed of a generator and a discriminator to optimize 2D and 3D linear elastic structures. Also, Zhang et al. [62] proposed a CNN model, consisting of an encoder and decoder, that predicts optimized designs in negligible time. Abueidda et al. [55] developed CNN models that predict the optimized structure while taking into account material and geometric nonlinearities as well as nonlinear stress constraints. Also, machine learning, such as the Kriging metamodel method, has been used to predict material microstructures' mechanical properties to accelerate multiscale topology optimization [63,64].

Most of the papers in the literature apply DL algorithms to accelerate the topology optimization process of structural systems. In the present paper, a single DL model, performing a real-time (non-iterative) topology optimization to optimize metamaterials with different desired properties, is presented. The data needed for training the model are generated using the energy-based homogenization method and periodic boundary conditions. This DL model optimizes for either bulk modulus (maximization), shear modulus (maximization), or Poisson's ratio (minimization). The remainder of the paper is laid out as follows: Section 2 presents an overview of the topology optimization problem along with the homogenization and periodic boundary conditions. Section 3 discusses the design space and training, validation, and testing datasets used. Section 4 scrutinizes the architecture of the CNN model, hyperparameters, and the loss function and metrics utilized in assessing the performance of the CNN model. In Section 5, we show the results along with the discussion. The study is concluded in Section 6 by summarizing the outcomes and highlighting potential future work.

2. Mathematical model

In addition to utilizing topology optimization algorithms to structural systems, they have been gaining an increasing interest in finding materials with optimal designs, leading to enhanced effective properties. To evaluate effective properties, one needs to use the homogenization theory and periodic boundary conditions (PBCs).

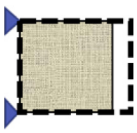
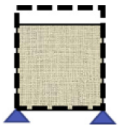
2.1. Homogenization and periodic boundary conditions

Within the realm of linear elasticity, the effective properties of periodic microstructures (cellular materials or composites) can be obtained using the homogenization method [65], using a single cell Y and applying PBCs. Topology optimization algorithms seek to identify the optimized design of that cell, called the representative unit cell (RUC), that maximizes or minimizes single or multiple objective function(s). The homogenized stiffness tensor E^H_{ijkl} of an RUC, with an area $|Y|$, is expressed as [23]:

$$\begin{aligned} E^H_{ijkl} &= \frac{1}{|Y|}\int_Y E_{pqrs}\,\varepsilon^{A(kl)}_{rs}\,\varepsilon^{A(ij)}_{pq}\,dY \\ \varepsilon^{A(ij)}_{pq} &= \varepsilon^{o(ij)}_{pq} - \varepsilon^{*(ij)}_{pq} \end{aligned} \tag{1}$$

where E_{pqrs} are the constitutive parameters, $\varepsilon^{o(kl)}_{pq}$ represents the three loading cases (for 2D) sketched in Table 1, and $\varepsilon^{*(kl)}_{pq}$ denotes the periodic solution of the variational problem. Then, one can combine the results of the three loading cases through a superposition to get the

Table 1
Sketch of the three loading cases used in homogenization.

			
$\varepsilon^{o(kl)}_{pq}$	$\varepsilon^{o(11)}_{11} = 1.0\%$ $\varepsilon^{o(11)}_{22} = 0$ $\varepsilon^{o(11)}_{12} = 0$	$\varepsilon^{o(22)}_{11} = 0$ $\varepsilon^{o(22)}_{22} = 1.0\%$ $\varepsilon^{o(22)}_{12} = 0$	$\varepsilon^{o(12)}_{11} = 0$ $\varepsilon^{o(12)}_{22} = 0$ $\varepsilon^{o(12)}_{12} = 1.0\%$

remaining components [66]. In finite element analysis, an RUC is discretized into N elements. Thus, Eq. (1) can be approximated by:

$$E^H_{ijkl} = \frac{1}{|Y|}\sum_{e=1}^{N}\left(\boldsymbol{u}_e^{A(ij)}\right)^T \boldsymbol{k}_e \boldsymbol{u}_e^{A(kl)} \tag{2}$$

where $\boldsymbol{k}_e$ is the element stiffness matrix, and $\boldsymbol{u}_e^{A(ij)}$ is the element displacement solution corresponding to $\varepsilon^{o(ij)}$.

Adopting Voigt notation for 2D, we write $11 \rightarrow 1, 22 \rightarrow 2$, and $12 \rightarrow 3$. Hence, the element stiffness can be expressed as:

$$\begin{bmatrix} E^H_{11} & E^H_{12} & E^H_{13} \\ E^H_{21} & E^H_{22} & E^H_{23} \\ E^H_{31} & E^H_{32} & E^H_{33} \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & Q_{13} \\ Q_{21} & Q_{22} & Q_{23} \\ Q_{31} & Q_{32} & Q_{33} \end{bmatrix} \tag{3}$$

where Q_{kl}, which is the sum of element mutual energies, is defined as:

$$Q_{kl} = \frac{1}{|Y|}\sum_{e=1}^{N}\left(\boldsymbol{u}_e^{A(k)}\right)^T \boldsymbol{k}_e \boldsymbol{u}_e^{A(l)}. \tag{4}$$

The strain fields $\varepsilon^{A(ij)}_{pq}$ shown in Eq. (1) are obtained by solving the equilibrium problem under the boundary conditions present in Table 1. Under the periodicity assumption, with a specified strain ε^o_{pq}, the displacement field is forced to satisfy the following condition [67]:

$$u_p^{i+} - u_p^{i-} = \varepsilon^o_{pq}\left(x_q^{i+} - x_q^{i-}\right) \tag{5}$$

where the superscripts $i+$ and $i-$ represent the pair of two opposite parallel boundary surfaces that are normal to the i-th direction.

2.2. Topology optimization

In this paper, the solid isotropic material penalization (SIMP) [68,69] approach is utilized to solve the optimization problem. The RUC is discretized into N finite elements, where each element has a corresponding density $\rho_e \in [0,1]$, which is a design variable. In the SIMP method, penalization factor $P = 5.0$ is defined and used to avoid intermediate values, i.e., to steer the densities to values of one or zero. The element Young's modulus E_e is expressed as:

$$E_e(\rho_e) = E_{min} + (E_o - E_{min})\rho_e^P \tag{6}$$

where E_o denotes the Young's modulus of a solid element ($\rho_e = 1$). On the other hand, E_{min} is defined as $E_{min} = 10^{-6}E_o$, representing the stiffness of a void element. Void elements are assigned stiffness values slightly higher than zero, as shown in Eq. (6), to avoid singularity of the stiffness matrix.

The element-based formulation, combined with first-order elements, encounters numerical instabilities such as mesh-dependence and checkerboarding [70,71]. Bruns et al. [72] used a density filtering technique to address those numerical instabilities. The pseudo-densities $\boldsymbol{\eta}$ are related to the physical (filtered) densities $\boldsymbol{\rho}$ as shown below:

$$\begin{aligned} \boldsymbol{\rho} &= \bar{\boldsymbol{W}}\boldsymbol{\eta} \\ w_{ij} &= \max(0, r_{min} - \|\boldsymbol{X}_i - \boldsymbol{X}_j\|^2) \\ \bar{w}_{ij} &= \frac{1}{\sum_{k=1}^{N_{r_{min}}} w_k} w_{ij} \end{aligned} \tag{7}$$

where r_{min} denotes the filter radius, w_{ij} represents the weight coefficient, $\overline{w}_{ij}$ is the normalized weight coefficient forming the normalized filter matrix $\overline{\boldsymbol{W}}$, and $\boldsymbol{X}_i$ denotes the coordinates of the centroid of element i. Note that $\boldsymbol{\eta}$ represents the design variables directly manipulated by the optimizer, while $\boldsymbol{\rho}$ is used to construct the global stiffness matrix based on Eq. (6).

The mathematical statement of the optimization problem is written as:

$$\begin{aligned} &\max_{\boldsymbol{\rho}} : f\left(E^H_{ijkl}(\boldsymbol{\rho})\right) \\ &\text{such that} : \boldsymbol{K}\boldsymbol{U}^{A(ij)} = \boldsymbol{F}^{A(ij)} \\ &\qquad\quad : V(\boldsymbol{\eta}) - V_f \le 0 \\ &\qquad\quad : 0 \le \rho_e \le 1 \end{aligned} \tag{8}$$

where $\boldsymbol{K}$ represents the global stiffness matrix, while $\boldsymbol{U}^{A(ij)}$ and $\boldsymbol{F}^{A(ij)}$ denote the global displacement and external force vectors, respectively, of the loading case (ij) (see Table 1). The volume fraction is defined by $V(\boldsymbol{\eta})$, where it is not allowed to exceed V_f. $V(\boldsymbol{\eta})$ is defined as:

$$V(\boldsymbol{\eta}) = \frac{\boldsymbol{\rho}^T \overline{\boldsymbol{V}}}{V_{total}} \tag{9}$$

where $\overline{\boldsymbol{V}}$ represents the vector of element volumes, and V_{total} is the total volume of the working domain. The objective function $f(E^H_{ijkl}(\boldsymbol{\rho}))$ is a functions of the homogenized stiffness tensor.

When the bulk modulus is maximized (for 2D) [13,73], $f(E^H_{ijkl}(\boldsymbol{\rho}))$ is expressed as:

$$f\left(E^H_{ijkl}(\boldsymbol{\rho})\right) = E_{1111} + E_{2222} + E_{1122} + E_{2211}. \tag{10}$$

For the case of shear modulus maximization (for 2D) [13,73], $f(E^H_{ijkl}(\boldsymbol{\rho}))$ is written as:

$$f\left(E^H_{ijkl}(\boldsymbol{\rho})\right) = E_{1212}. \tag{11}$$

Constructing materials with a negative Poisson's ratio is a challenging task, as it requires imposing other constraints such as bulk modulus and/or isotropy. For further details, readers are referred to previous works [15,23,73–76]. Here, $f(E^H_{ijkl}(\boldsymbol{\rho}))$ is written as:

$$f\left(E^H_{ijkl}(\boldsymbol{\rho})\right) = \alpha^a(E_{1111} + E_{2222}) - E_{1122} \tag{12}$$

where $\alpha \in (0,1)$ is a relaxation parameter, and, in this study, $\alpha = 0.75$. a is the iteration number. As the optimization iterations progress, the first term $\alpha^a(E_{1111} + E_{2222})$ vanishes, while the second term gets maximized, i.e., E_{1122} gets minimized. The sensitivities of the different objective functions are calculated using the adjoint method, where the sensitivities of E^H_{ijkl} with respect to the densities $\boldsymbol{\rho}$ is expressed as:

$$\frac{\partial E^H_{ijkl}}{\partial \rho_e} = \frac{1}{|Y|}P\rho_e^{P-1}(E_o - E_{min})\left(\boldsymbol{u}_e^{A(ij)}\right)^T \boldsymbol{k}_o \boldsymbol{u}_e^{A(kl)} \tag{13}$$

where $\boldsymbol{k_o}$ is the element stiffness matrix when $\rho_e = 1$.

3. Data description, generation, and processing

A high-level flowchart of the data preparation process is given in Fig. 1. We computationally generated a pool of data. We start from a topology optimization problem with randomly sampled optimization parameters. The problem has three optimization parameters: a filter

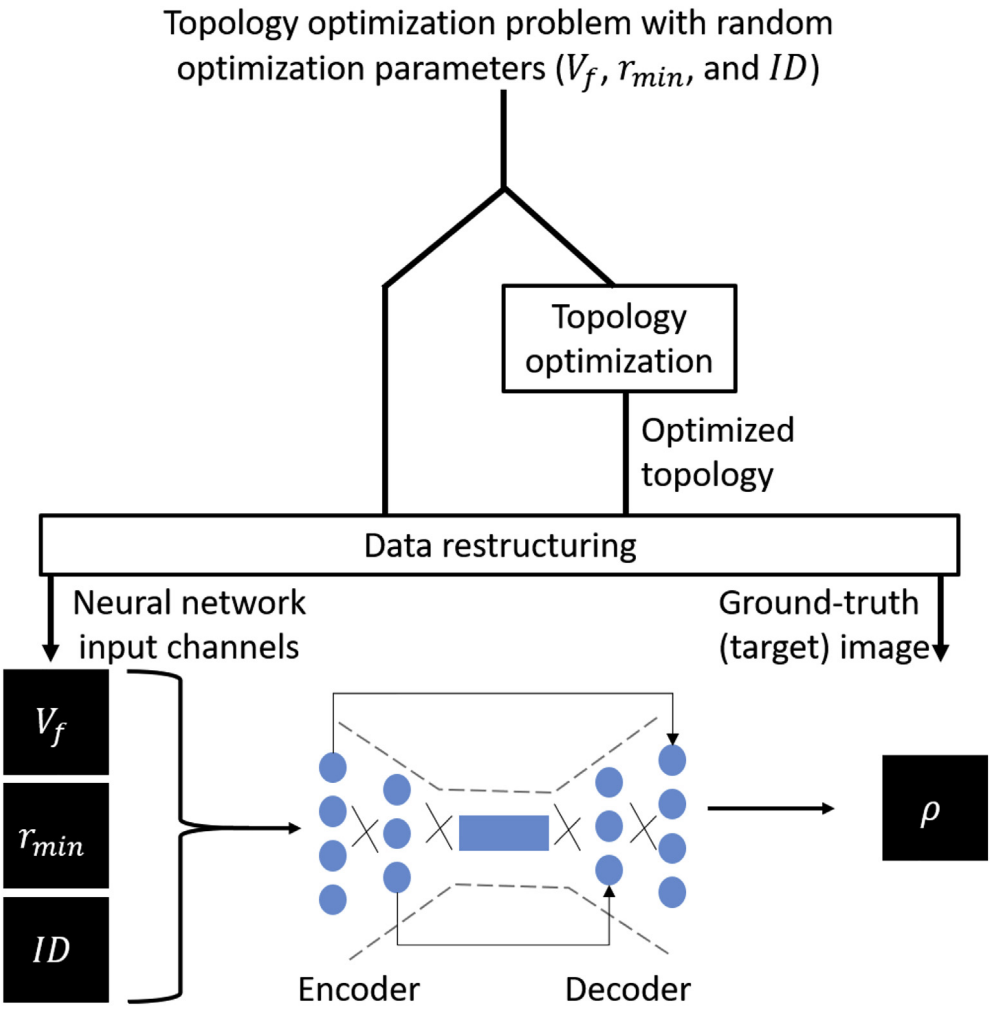


Fig. 1. Computational pipeline for the preparation of data.

radius, a design constraint (volume fraction), and a design objective (maximum bulk modulus, maximum shear modulus, or minimum Poisson's ratio). The size of the RUC is 128μm $\times$ 128μm, while we randomly sample the volume fraction ($V_f \in [0.2, 0.8]$) and the filter radius ($r_{min} \in [2.0, 10.0]\mu$m) from uniform distributions. Boundary conditions, needed to define a well-posed optimization problem, are shown in Table 1. The optimized topology is found by utilizing a conventional optimization approach, as outlined in Section 2.

The CNN model takes three input images (one for each optimization parameter) and produces an output image. In other words, each datapoint consists of three input images and the corresponding ground-truth (target) image. All images (channels) have a 128 $\times$ 128 pixels, corresponding to the 128 $\times$ 128 finite element discretization of the physical domain. For the volume fraction and the filter radius images, all pixels are assigned a value sampled from the corresponding ranges. For the design objective image, all pixels are assigned a numeric identifier $ID \in \{0,1,2\}$, where $ID = 0$ represents maximizing of the bulk modulus, $ID = 1$ represents maximizing the shear modulus, and $ID = 2$ represents minimizing the Poisson's ratio. Note that one may generally choose a different set of identifiers as long as the labeling consistency is maintained. The target image represents the optimized topology. The pixels' values on the target image are the densities obtained from the topology optimization algorithm. Note that although the input parameters are three constant parameters, they are used to construct three images (channels). This allows us to use convolutional neural networks, which have been proven successful in obtaining outputs in the form of images, while maintaining the generalizability of the CNN model, as discussed in Section 5.

We generate a pool of 30,000 datapoints. The size of the data pool is determined by the performance of the neural network. The convergence condition for setting the size of the generated dataset is a dice similarity coefficient (DSC) higher than 0.95. The interpretation of the DSC is discussed in more detail in Section 4. Numerical algorithms for all data generation steps (initial randomization, topology optimization, and data restructuring) are coded by leveraging the MATLAB software's programming environment. High-throughput computations are performed on iForge; an HPC cluster hosted at the National Center for Supercomputing Applications (NCSA). A representative compute node on iForge has the following hardware specifications: two 20-core Intel Skylake CPUs, two NVIDIA Tesla V100 GPUs, and 192 GB main memory.

4. Deep learning model

4.1. Architecture

The CNN model in this paper is based on the ResUNet [77] (see Fig. 2a). This is a semantic segmentation neural network, which takes advantage of residual learning and U-Net. By combining these two, it is possible to use the strengths of both. The network is built with residual units but uses a similar architecture to U-Net [77]. We use this CNN model to find optimized material distributions of 2D architected

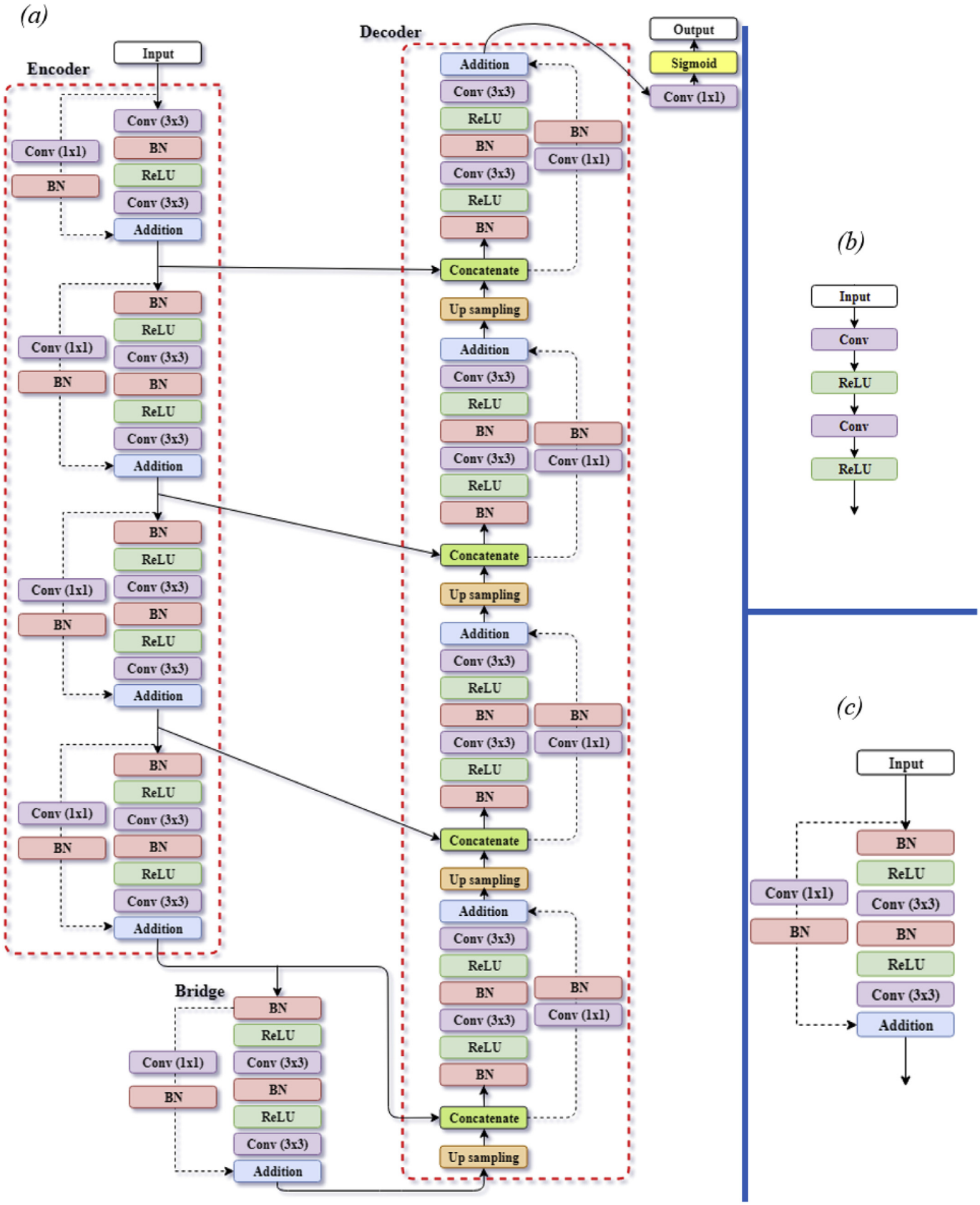


Fig. 2. Illustration of the a) architecture of the ResUNet, b) building block used in U-Net, and c) building block used in ResUNet.

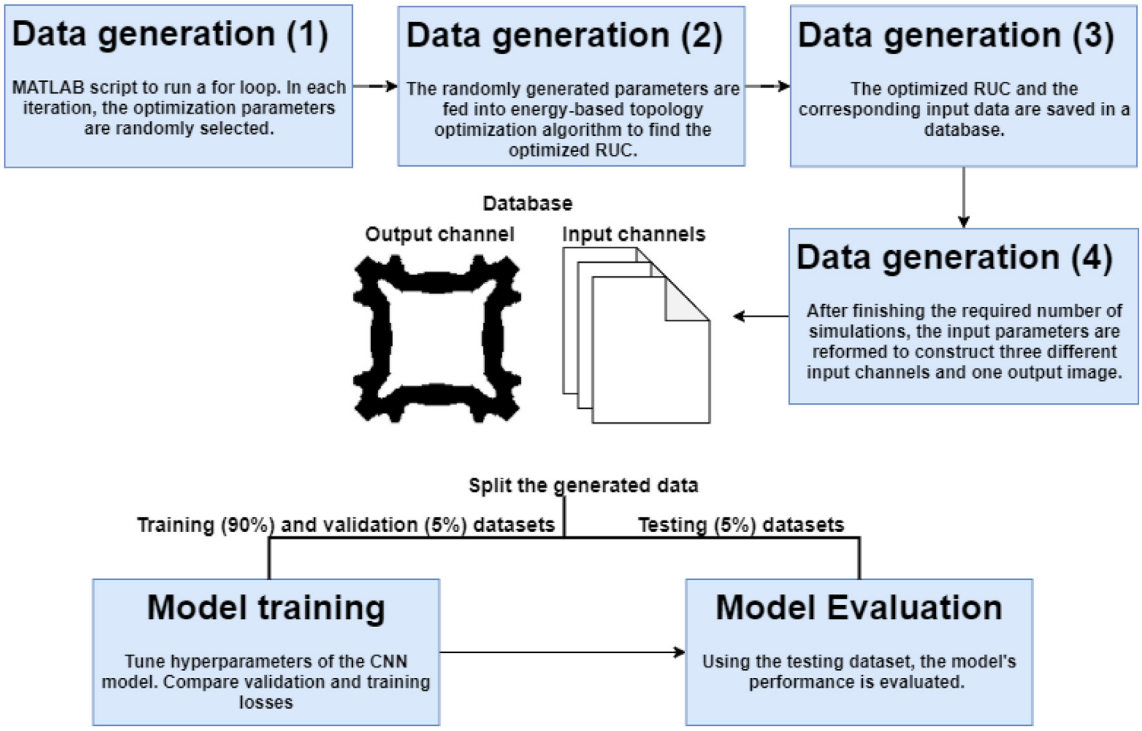


Fig. 3. Flowchart showing the different steps used to develop a CNN-based optimizer.

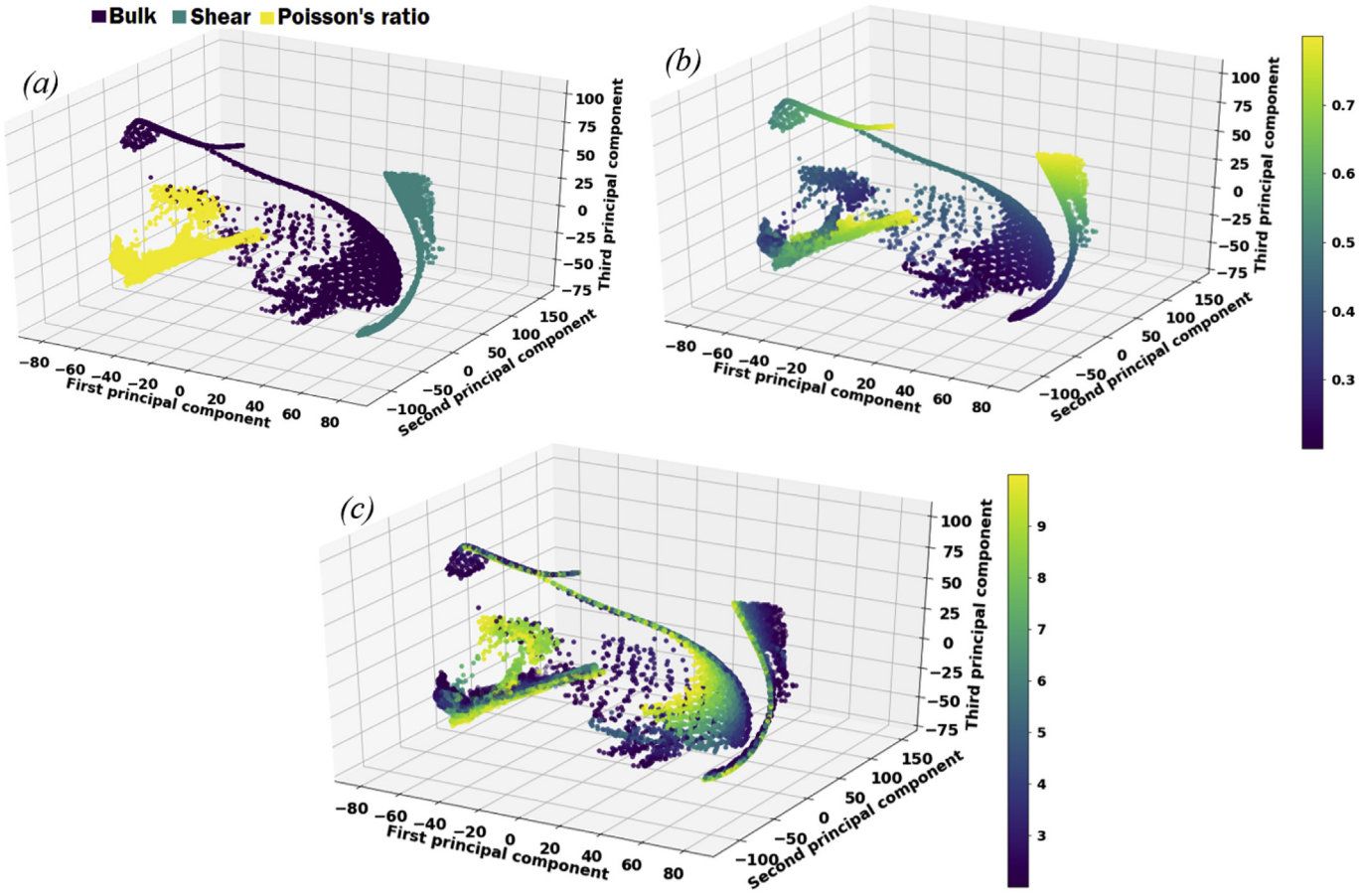


Fig. 4. Principal component analysis (PCA) for the generated data: a) design objective (*ID*), b) volume fraction (V_f), and c) filter radius (r_{min} in μm). The optimizer maximizes the bulk and shear moduli while it minimizes the Poisson's ratio.

materials trained using topology optimization data. U-Net, first proposed by Ronneberger et al. [78], improves segmentation accuracy by concatenating feature maps from different neural network levels. The main difference between the U-Net and the ResUNet, used here, is that the ResUNet uses residual units instead of neural units as the building block. Fig. 2b and c show the building block used in the conventional U-Net and the ResUNet, respectively. As shown in Fig. 2c, ResUNet uses batch normalizations (BN), rectified linear units (ReLU), and convolutional layers (Conv), whereas shown in Fig. 2b, U-Net uses ReLU and convolutional layers in the building block.

Fig. 2a shows the architecture of the employed ResUNet. The ResUNet can be broken into three main parts. The first part, the encoder, has the task of encoding (compressing) input images into compact representation (low-dimensional latent space) by a series of convolutional layers. The second part, the decoder, receives the encoded input images from the bridge and constructs (by mapping to targets) the full representative unit cell (RUC). The third part, the bridge, connects the encoder and the decoder. In other words, the low-dimensional latent space forms a bottleneck, which forces the encoder to evoke the critical features from the input channels and represent them in a meaningful

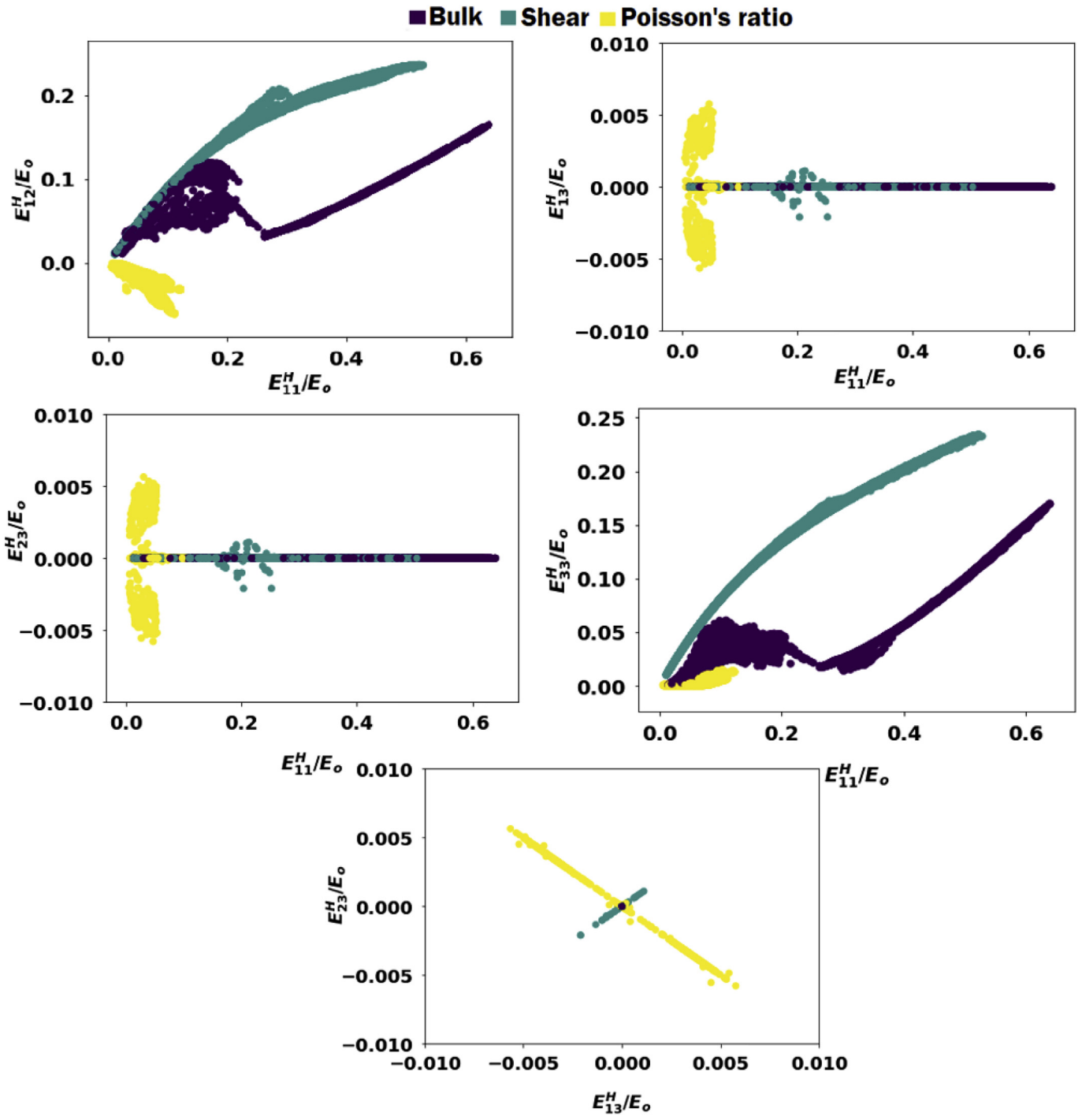


Fig. 5. Property distribution space of metamaterials optimized for different objectives. Property distribution space of metamaterials optimized for different objectives. The different colors represent the properties optimized for. The optimizer maximizes the bulk and shear moduli while it minimizes the Poisson's ratio.

embedding. The skip connections between the decoder and encoder and within the residual units facilitate information propagations in forward and backward directions and lessen the parameters needed. For more information, the paper by Zhang et al. [77] details the architecture of the ResUNet and highlights its strengths compared to other neural network models.

4.2. Loss function and model evaluation

The CNN model is developed and tested using Keras [79] with a TensorFlow [80] backend. The hyperparameters used in the CNN model are a batch size of 128 and a learning rate of 0.001. We used Adam optimizer [81], which is an algorithm for gradient-based stochastic optimization. Also, mini-batching is used to improve the convergence rate and help the CNN model not be trapped in local minima [82]. The optimizer and hyperparameters are selected after a series of testing of the CNN model's performance such that the loss function is minimized and the DSC (discussed below) is maximized.

The training process is an optimization problem, where the weights $\boldsymbol{W}$ and biases $\boldsymbol{b}$ of the CNN model are found such that the loss (mean square error) between the segmentation generated by the CNN model $\boldsymbol{O}(\boldsymbol{I};\boldsymbol{W},\boldsymbol{b})$ and the ground-truth segmentation $\boldsymbol{T}(\boldsymbol{I})$ is minimized. $\boldsymbol{I}$ denotes the input channels. The mean square error (MSE) is mathematically expressed as:

$$\text{MSE} = \frac{1}{M}\sum_{i=1}^{M} \|\boldsymbol{T}_i(\boldsymbol{I}_i) - \boldsymbol{O}_i\boldsymbol{I}_i;\boldsymbol{W},\boldsymbol{b}\|^2 \tag{14}$$

where M is the total number of training datapoints. We also measure the dice similarity coefficient (DSC) [83] alongside the MSE in order to measure the performance by comparing the ground-truth images and the predicted ones by the CNN. The DSC measures the similarity between two images, and here, we evaluate the performance of the CNN model by monitoring the mean DSC, which is written as:

$$\text{DSC} = \frac{1}{M}\sum_{i=1}^{M} \frac{2\,|\,\boldsymbol{O}_i\boldsymbol{I}_i;\boldsymbol{W},\boldsymbol{b} \cap \boldsymbol{T}_i(\boldsymbol{I}_i)\,|}{|\,\boldsymbol{O}_i\boldsymbol{I}_i;\boldsymbol{W},\boldsymbol{b}\,| + |\,\boldsymbol{T}_i(\boldsymbol{I}_i)\,|}. \tag{15}$$

If the two images are the same, meaning the prediction matched the ground-truth image perfectly, then the DSC would be equal to 1.0. If there are no common pixels, then the DSC will measure to be 0.0. Hence, the higher the mean DSC, the better accuracy is expected from the CNN model. Note that the DSC is not directly maximized, but we minimize the MSE (defined in Eq. (14)), and then the DSC (defined in Eq. (15)) is calculated and used as an independent metric to evaluate the performance of the model. Fig. 3 summarizes the different stages of the development of the CNN model.

5. Results and discussion

One way for visualizing a large pool of datapoints is to use PCA, which reduces the dimensionality to increase the interpretability of the space. Before performing PCA, all targets (output images) are scaled and transformed, $z = \frac{\rho-\mu}{s}$ where μ is the mean of the datapoints, and s is their standard deviation. We find the first three principal components, and by calculating the variance ratio of the principal components, we find that the first three components contribute 51% of the total variance. Fig. 4 portrays the distribution of the datapoints in the space defined by the first three principal components of the densities of the generated datapoints, where the scale bars represent the *ID* in Fig. 4a, the volume fraction in Fig. 4b, and the filter radius in Fig. 4c. Fig. 4a indicates that there is a clear distinction between the targets optimized for different objective designs. In other words, each design objective occupies (or is clustered in) a different region in the output space.

Fig. 5 portrays the property distribution space of the components of the homogenized stiffness tensor. The components shown in Fig. 5 are represented in Voigt notation (see Eq. (3)), and they are normalized by the Young's modulus of the base material E_o (see Eq. (6)). The generated microstructures possess a level of symmetry due to the nature of the applied strain and the used initial design (initial guess), specifically, $E^H_{1111} = E^H_{2222}$ (or in Voigt notation $E^H_{11} = E^H_{22}$). The effect of the initial design is discussed below in more detail. Also, due to the symmetry of the stress and strain tensors, $E^H_{12} = E^H_{21}$, $E^H_{13} = E^H_{31}$, and $E^H_{23} = E^H_{32}$. When the bulk modulus is maximized, the optimizer tends to maximize E^H_{11} and E^H_{12} simultaneously. In the case of maximizing the shear modulus, the optimizer yields solutions with high E^H_{33} values. When the Poisson's ratio is minimized, the converged solutions exhibit low positive values of E^H_{11} and negative values of E^H_{12}. This indicates that the optimizer generally manages to produce metamaterials with negative Poisson's ratio, such materials are usually called auxetic materials in the literature, without losing its load-bearing capability ($E^H_{11} > 0$). For the cases of maximizing bulk and shear moduli, the converged solutions have almost zero values of E^H_{13} and E^H_{23}, i.e., $E^H_{13} \approx E^H_{23} \approx 0.0$. However, when the Poisson's ratio is minimized, $E^H_{13} \approx -E^H_{23}$.

We start by considering different training dataset sizes. As mentioned earlier, 30,000 datapoints were generated. We use 20%, 40%, 60%, 80%, and 90% of the generated data to train the model, and we report the mean square errors and mean dice similarity coefficient in Fig. 6. One can infer that adding more data to the training dataset will not significantly enhance the CNN model's performance, and a 90% split deems acceptable. The results reported below are for the 90% split. Specifically, the data generated was split into three different datasets for different uses. 90% of the generated data is for training, 5% is for validation, and 5% is for testing. The training dataset is used to obtain the CNN model's weights and biases through solving the optimization problem (minimizing the loss function). The validation data are utilized to evaluate the convergence of the CNN model and not used to find the optimized parameters of the CNN model. More specifically, after each epoch, the losses obtained from the validation and training datasets are compared to recognize issues with the training process, such as overfitting or underfitting. Lastly, the testing dataset is used as a sanity check, i.e., check whether the trained model generalizes to datapoints previously unseen.

The model is trained using 150 epochs. The training time is 9.1 hours. Fig. 7a and b show the convergence history of the loss function and DSC for the CNN model, respectively. The difference between the validation and training losses is small, as depicted in Fig. 7a. The training and validation DSCs are also quite high (close to 1.0), as shown in Fig. 7b. These observations about the training and validation losses and DSCs indicate there is no significant overfitting or underfitting. To evaluate the model's performance further, we look at the mean square error and

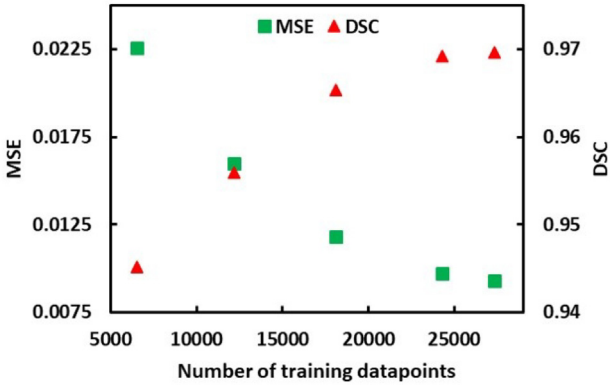


Fig. 6. Variation of the mean square error and mean dice similarity coefficient with the training dataset size.

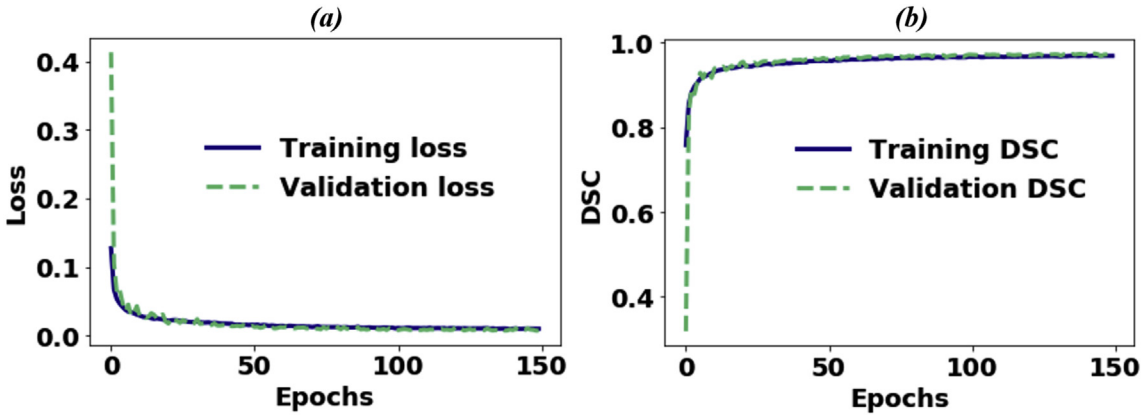


Fig. 7. The convergence history of the a) mean square error and b) mean dice similarity coefficient for the CNN model.

mean DSC calculated using the testing dataset, after the training process is completed. Note that the testing dataset was never seen by the model during the training process. The testing mean square error and mean DSC are 0.00794 and 0.970, respectively. Also, the objective function and the volume of the optimized RUCs obtained from the CNN model are compared with those of the ground-truth. The mean deviations of the objective function and volume are 2.9% and 1.2%, respectively. Fig. 8 shows three examples obtained from the testing dataset, without using a binarization scheme. For the case presented in Fig. 8a, the deviations of the prediction objective function and volume from those of the

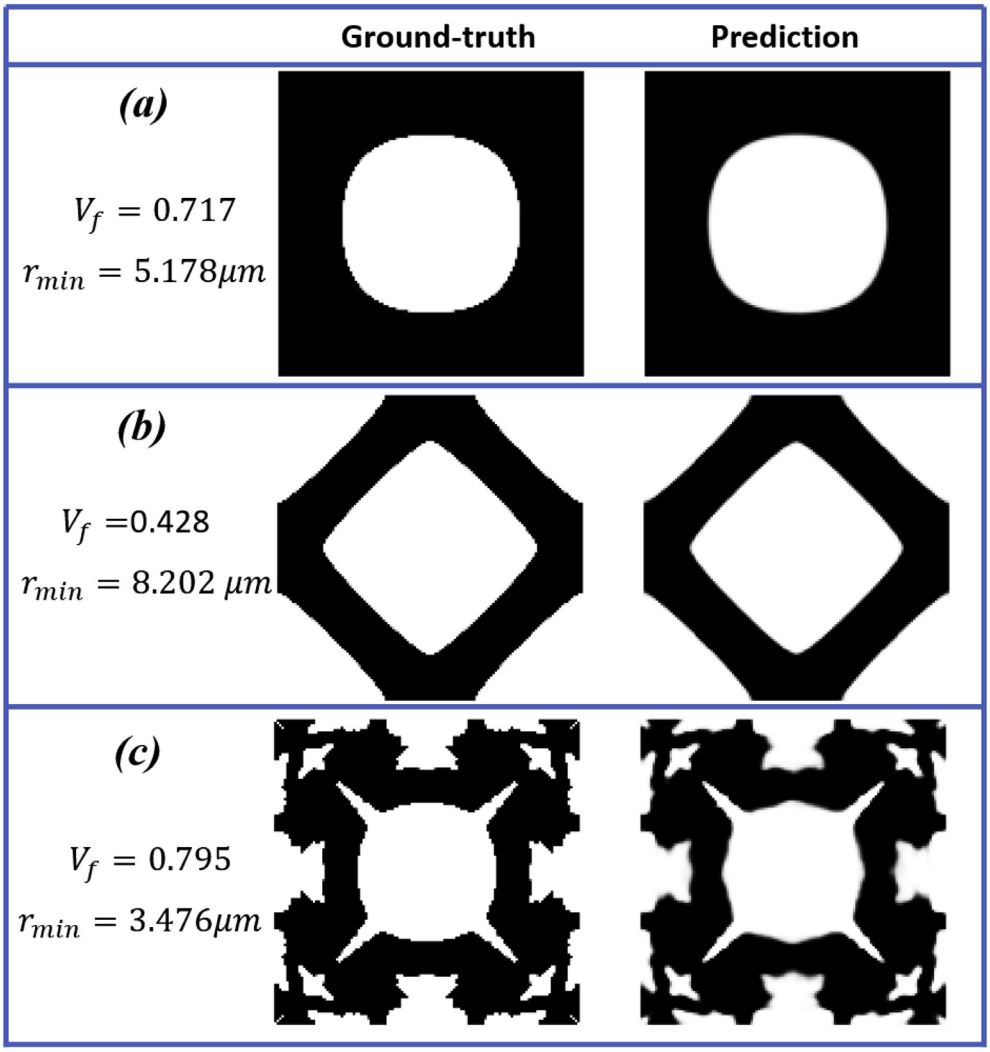


Fig. 8. Comparison between optimized designs for the case of a) maximizing the bulk modulus, b) maximizing the shear modulus, and c) minimizing the Poisson's ratio. No binarization scheme is used.

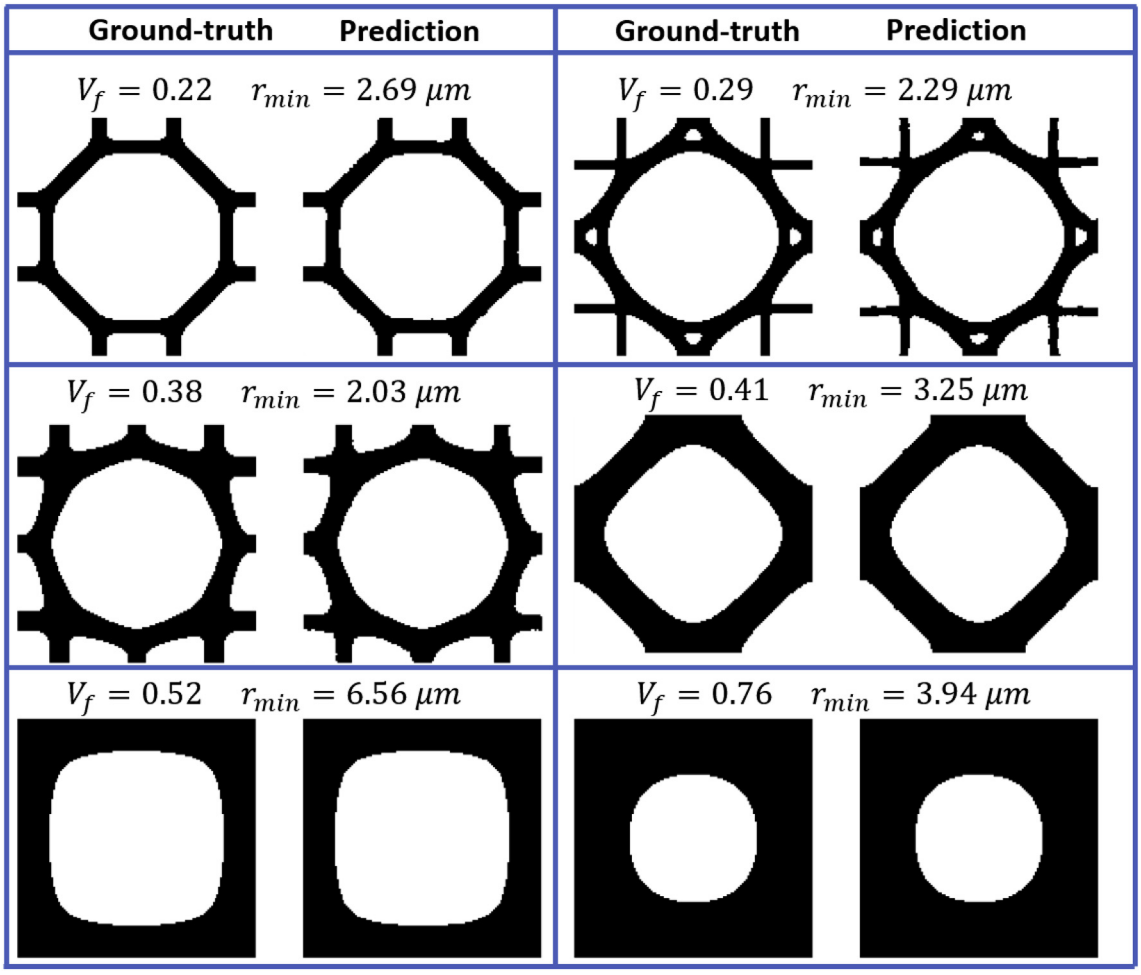


Fig. 9. Comparison between optimized designs for the case of maximizing the bulk modulus.

ground-truth are 2.16% and 0.16%, respectively. For the example presented in Fig. 8b the deviations of the prediction objective function and volume from those of the ground-truth are 0.74% and 1.49%, respectively. For the case presented in Fig. 8c, the deviations of the prediction objective function and volume from those of the ground-truth are 3.29% and 1.06%, respectively. These several indicators imply that the ground-truth and the predicted images are very similar.

The densities of the optimized designs range from 0 to 1. For practical purposes and manufacturability, one needs to decide whether a material should exist for a specific element. Thus, we used a threshold value of 0.5 in order to retrieve the binary nature of the problem. Elements with densities higher than or equal to 0.5 are set to 1, and elements with densities less than 0.5 are set to 0. Figs. 9, 10, and 11 show comparisons between the ground-truth and predicted images for the bulk modulus (maximization), shear modulus (maximization), and Poisson's ratio (minimization), respectively. These examples are randomly selected from the testing dataset, unseen by the model throughout the training process, to perform a qualitative evaluation for the model's performance. Then, they are grouped according to the design objective, as laid out in Figs. 9, 10, and 11. The predicted distributions are near identical to the ground-truth. Also, note that the filtering technique discussed in Section 2.2 prevented checkerboard patterns from taking place. Several data-driven models available in the literature encountered disconnections in the some of the obtained optimal designs, while such disconnections did not appear in the ground-truth data [45,58]. Although we are using a much larger design domain 128×128, such an issue was not observed due to the inherent nature of the model architecture used. To improve the quality of the results obtained from the proposed CNN model, one can add one or more residual blocks as well as generate more data for training and testing the model.

For density-based topology optimization, it is common to use an initial design for the working domain with a homogeneous distribution (uniform densities). More specifically, all elements in the initial design are assigned a density value equal to the volume fraction constraint. This is generally useful to avoid any predefined geometries leading to steering the optimizer into a particular local optimum. Since, for structural design optimization, the displacement field is non-uniform in the domain because of the prescribed boundary condition, such a strategy is successful. The non-uniform displacement field yields non-uniform sensitivities, which the optimizer uses to obtain the next design point in the design space. However, when the homogenization theory is used, the RUC as a whole is deformed by the macroscopic strain tensor. Hence, an initial design with a uniform density distribution leads to a homogeneous microscale displacement field and uniform sensitivities. Consequently, the optimizer will fail to improve the design. Many researchers suggested using initial designs with non-uniform density distributions to induce non-uniform sensitivities [56,84,85]. Zhang et al. [86] proposed an initialization scheme that uses up to six initial material layouts and compares the optimized solutions.

In this study, we define a circular region at the center of the working domain, possessing lower initial densities, while keeping this region fixed for all datapoints. Note that this initial design is one of the designs considered by Zhang et al. [86]. Such a strategy biases the optimizer toward a specific design. Such an issue is not paralyzing such that it

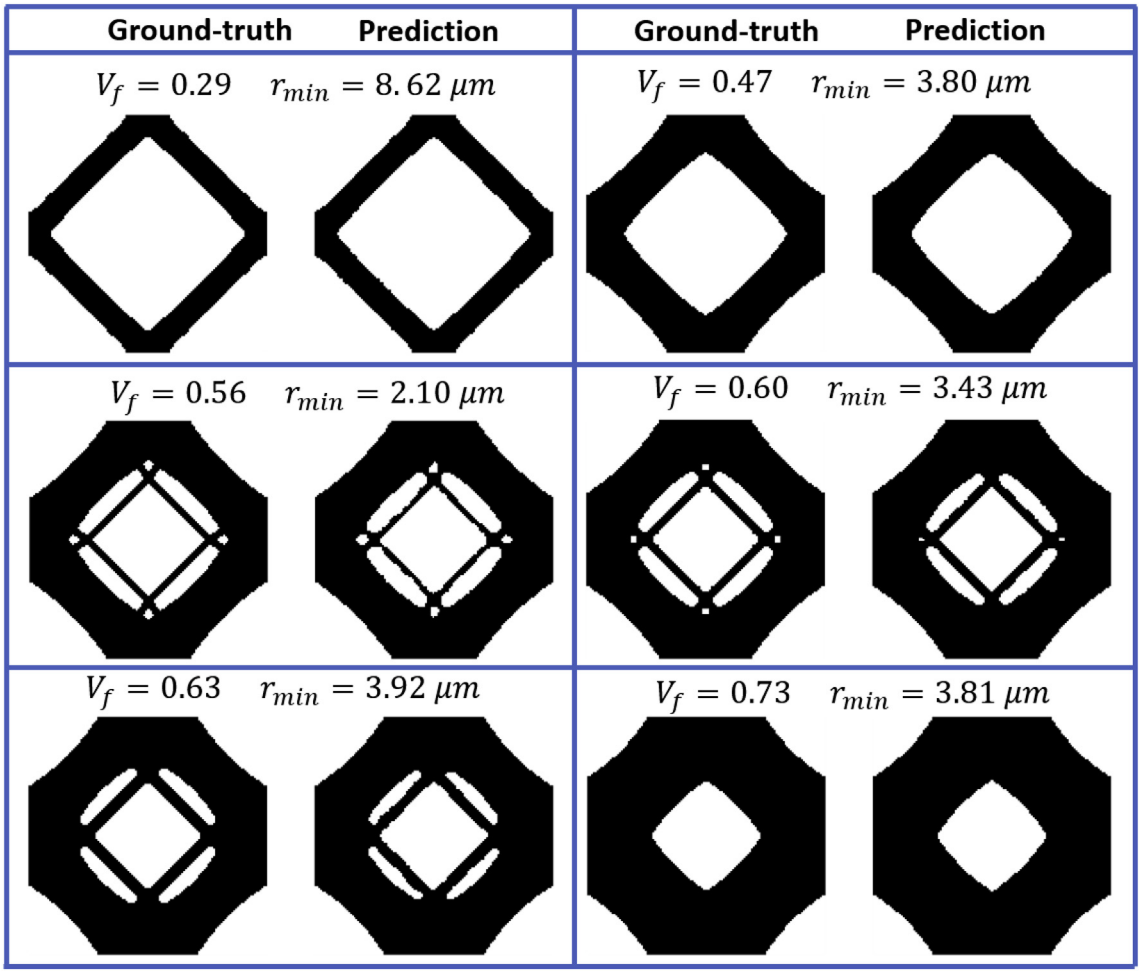


Fig. 10. Comparison between optimized designs for the case of maximizing the shear modulus.

prevents researchers from using topology optimization algorithms. However, one needs to be aware of it. One method that can be used to mitigate such an effect is to use multiple initial designs and compare the performance of the different designs. This adds another level of complexity to the optimization problem of metamaterials. CNN models can be convenient and accelerate the process of obtaining optimized designs. One advantage of having the input parameters as three different channels (see Fig. 1) rather than constants is that it maintains the generalizability of the developed CNN model. For instance, one can add an extra channel to the existing input channels (see Fig. 1). The values of this channel will be the initial densities of the elements. Then, one can get optimized topologies with different initial designs almost instantly. A similar strategy can also be adopted to account for passive elements, which are elements with a density value of zero. In other words, the densities of passive elements are known in a priori; hence, they are not design variables (check Sigmund's work [87] for further details about passive elements). One can add an another channel defining active ($\rho_e = 1.0$) and passive ($\rho_e = 0.0$) elements to the existing ones (see Fig. 1).

To obtain an optimized microstructure using conventional topology optimization algorithms, one needs many optimization iterations. Each optimization iteration requires solving three finite element problems (illustrated in Table 1) and then calculating the sensitivities. However, after having the CNN model trained, validated, and tested, the model with its learnable parameters (weights and biases) can be moved to any low-end computing platform such as a laptop. The optimized solutions are found there almost instantly (non-iteratively) for any variation of input parameters. One potential application for the proposed model is to produce almost instantly preliminary optimized designs. Specifically, such a model can yield near-optimal initial designs that can be used for subsequent conventional topology optimization and significantly increasing convergence rate and computational and overall efficiencies of the optimization process. Another use for such surrogate data-driven models is to use them as building blocks for the design and optimization of multiscale metamaterial systems. The desired performance of the entire multiscale metamaterial structure will be improved [32,88,89], while a much higher computational efficiency will be achieved by replacing nested microscale topology optimization calculations for every unit cell by an instant inference from a properly trained neural network. Hence, such data-driven topology optimization models can be a great complement to the existing topology optimization algorithms.

Finally, while we have demonstrated in this work how the deep learning methods can assist metamaterial designs in 2D, conceptually, nothing will be different in 3D, except that the computational cost for generating training data from 3D topological optimizations will be much higher. However, FEA solvers and partitioning methods, which consume most of the computational time in large-scale 3D topology optimization frameworks, have shown an exceedingly improved parallel performance on modern high-performance computing [90,91]. Moreover, once the sufficient 3D training data is generated and properly trained on a high-end computing system, topology optimization of 3D RUCs will still be inferenced almost instantly even on low-end computers, which may make this and other similar data-driven methods an essential tool for quick optimization of metamaterials.

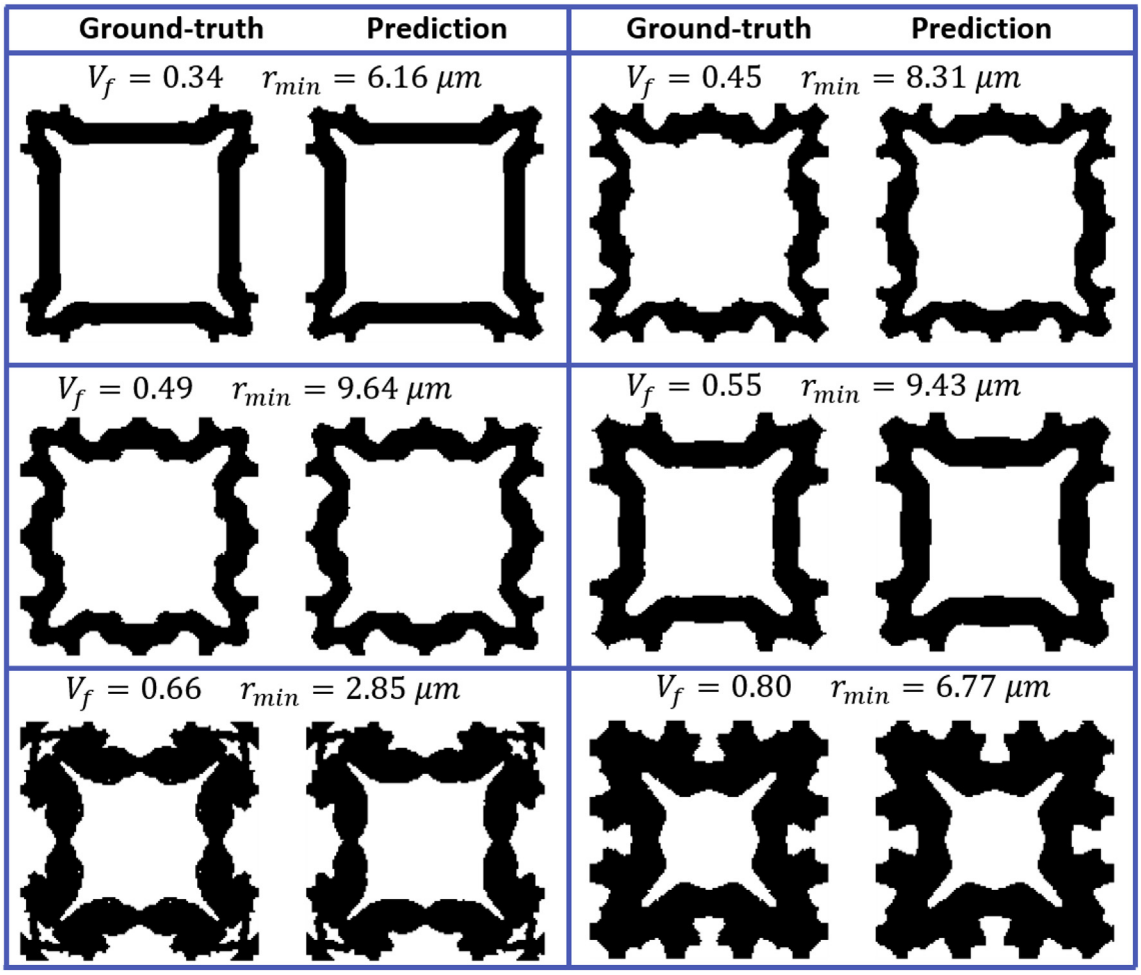


Fig. 11. Comparison between optimized designs for the case of minimizing the Poisson's ratio.

6. Conclusions

In this paper, we develop a CNN model in order to predict the optimized metamaterial design for either maximizing the bulk modulus, maximizing the shear modulus, or minimizing the Poisson's ratio. A topology optimization framework, involving the energy-based homogenization method and periodic boundary conditions, is used to generate the data required for training the CNN model. The model developed has shown to be robust in terms of accuracy and inference time. Using this CNN model, we are able to almost instantly have quality topology optimization results on a low-end computing platform, such as laptops. Such a model also opens the door for faster topology optimization algorithms for multiscale metamaterial systems and other computationally expensive topology optimization problems.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors would like to thank the National Center for Supercomputing Applications (NCSA) Industry Program and the Center for Artificial Intelligence Innovation.

References

[1] X. Yu, J. Zhou, H. Liang, Z. Jiang, L. Wu, Mechanical metamaterials associated with stiffness, rigidity and compressibility: a brief review, Prog. Mater. Sci. 94 (2018) 114–173.
[2] L.J. Gibson, M.F. Ashby, Cellular Solids: Structure and Properties, Cambridge University Press, 1999.
[3] A.M. Abou-Ali, O. Al-Ketan, R. Rowshan, R. Abu Al-Rub, Mechanical response of 3d printed bending-dominated ligament-based triply periodic cellular polymeric solids, J. Mater. Eng. Perform. 28 (4) (2019) 2316–2326.
[4] D.W. Abueidda, M. Elhebeary, C.-S.A. Shiang, R.K.A. Al-Rub, I.M. Jasiuk, Compression and buckling of microarchitectured neovius-lattice, Extreme Mech. Lett. (2020) 100688.
[5] M.F. Ashby, H. Shercliff, D. Cebon, Materials: Engineering, Science, Processing and Design, Butterworth-Heinemann, 2018.
[6] D.W. Abueidda, P. Karimi, J.-M. Jin, N.A. Sobh, I.M. Jasiuk, M. Ostoja-Starzewski, Shielding effectiveness and bandgaps of interpenetrating phase composites based on the schwarz primitive surface, J. Appl. Phys. 124 (17) (2018) 175102.
[7] A.S. Dalaq, F. Barthelat, Three-dimensional laser engraving for fabrication of tough glass-based bioinspired materials, JOM (2020) 1–11.
[8] A.S. Dalaq, F. Barthelat, Strength and stability in architectured spine-like segmented structures, Int. J. Solids Struct. 171 (2019) 146–157.
[9] L. Valdevit, J. Bauer, Fabrication of 3d micro-/nanoarchitected materials, Three-dimensional Microfabrication Using Two-photon Polymerization, Elsevier 2020, pp. 541–576.

[10] M. Ruzzene, F. Scarpa, Directional and band-gap behavior of periodic auxetic lattices, Phys. Status Solidi B 242 (3) (2005) 665–680.
[11] Z. Vangelatos, G.X. Gu, C.P. Grigoropoulos, Architected metamaterials with tailored 3d buckling mechanisms at the microscale, Extreme Mech. Lett. 33 (2019) 100580.
[12] U.G. Wegst, H. Bai, E. Saiz, A.P. Tomsia, R.O. Ritchie, Bioinspired structural materials, Nat. Mater. 14 (1) (2015) 23–36.
[13] J. Gao, H. Li, L. Gao, M. Xiao, Topological shape optimization of 3d micro-structured materials using energy-based homogenization method, Adv. Eng. Softw. 116 (2018) 89–102.
[14] A.S. Dalaq, F. Barthelat, et al., Mater. Des. (2020) 108889.
[15] M.P. Bendsoe, O. Sigmund, Topology Optimization: Theory, Methods, and Applications, Springer Science & Business Media, 2013.
[16] M.P. Bendsoe, N. Kikuchi, Generating optimal topologies in structural design using a homogenization method, Comput. Methods Appl. Mech. Eng. 71 (1988) 197–224.
[17] H. Bikas, P. Stavropoulos, G. Chryssolouris, Additive manufacturing methods and modelling approaches: a critical review, Int. J. Adv. Manuf. Technol. 83 (1–4) (2016) 389–405.
[18] B.H. Jared, M.A. Aguilo, L.L. Beghini, B.L. Boyce, B.W. Clark, A. Cook, B.J. Kaehr, J. Robbins, Additive manufacturing: toward holistic design, Scr. Mater. 135 (2017) 141–147.
[19] J. Gardan, Additive manufacturing technologies: state of the art and trends, Int. J. Prod. Res. 54 (10) (2016) 3118–3132.
[20] O. Al-Ketan, D.-W. Lee, R. Rowshan, R.K.A. Al-Rub, Functionally graded and multi-morphology sheet tpms lattices: design, manufacturing, and mechanical properties, J. Mech. Behav. Biomed. Mater. 102 (2020) 103520.
[21] A. Alhammadi, O. Al-Ketan, K.A. Khan, M. Ali, R. Rowshan, R.K.A. Al-Rub, Microstructural characterization and thermomechanical behavior of additively manufactured alsi10mg sheet cellular materials, Mater. Sci. Eng. A 139714 (2020).
[22] M. Osanov, J.K. Guest, Topology optimization for architected materials design, Annu. Rev. Mater. Res. 46 (2016) 211–233.
[23] O. Sigmund, Materials with prescribed constitutive parameters: an inverse homogenization problem, Int. J. Solids Struct. 31 (17) (1994) 2313–2329.
[24] O. Sigmund, Tailoring materials with prescribed elastic properties, Mech. Mater. 20 (4) (1995) 351–368.
[25] M. Neves, H. Rodrigues, J.M. Guedes, Optimal design of periodic linear elastic microstructures, Comput. Struct. 76 (1–3) (2000) 421–429.
[26] X. Huang, A. Radman, Y.M. Xie, Topological design of microstructures of cellular materials for maximum bulk or shear modulus, Comput. Mater. Sci. 50 (6) (2011) 1861–1870.
[27] H. Zhang, Y. Luo, Z. Kang, Bi-material microstructural design of chiral auxetic metamaterials using topology optimization, Compos. Struct. 195 (2018) 232–248.
[28] H. Li, Z. Luo, L. Gao, P. Walker, Topology optimization for functionally graded cellular composites with metamaterials by level sets, Comput. Methods Appl. Mech. Eng. 328 (2018) 340–364.
[29] H. Zong, H. Zhang, Y. Wang, M.Y. Wang, J.Y. Fuh, On two-step design of microstructure with desired poisson's ratio for am, Mater. Des. 159 (2018) 90–102.
[30] M. Ye, L. Gao, H. Li, A design framework for gradually stiffer mechanical metamaterial induced by negative poisson's ratio property, Mater. Des. 108751 (2020).
[31] P. Vogiatzis, S. Chen, X. Wang, T. Li, L. Wang, Topology optimization of multi-material negative poisson's ratio metamaterials using a reconciled level set method, Comput. Aided Des. 83 (2017) 15–32.
[32] D.L. McDowell, J. Panchal, H.-J. Choi, C. Seepersad, J. Allen, F. Mistree, Integrated Design of Multiscale, Multifunctional Materials and Products, Butterworth-Heinemann, 2009.
[33] W. Lim, D. Jang, T. Lee, Speech emotion recognition using convolutional and recurrent neural networks, 2016 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA), IEEE 2016, pp. 1–4.
[34] Z. Thorat, S. Mahadik, S. Mane, S. Mohite, A. Udugade, Self Driving Car Using Raspberry-pi and Machine Learning 6 (2019) 969–972.
[35] M. de Bruijne, Machine learning approaches in medical image analysis: from detection to diagnosis, Med. Image Anal. 33 (2016) 94–97.
[36] D.W. Abueidda, M. Almasri, R. Ammourah, U. Ravaioli, I.M. Jasiuk, N.A. Sobh, Prediction and optimization of mechanical properties of composites using convolutional neural networks, Compos. Struct. 227 (2019) 111264.
[37] C. Yang, Y. Kim, S. Ryu, G.X. Gu, Using convolutional neural networks to predict composite properties beyond the elastic limit, MRS Commun. 9 (2) (2019) 609–617.
[38] M. Mozaffar, R. Bostanabad, W. Chen, K. Ehmann, J. Cao, M. Bessa, Deep learning predicts path-dependent plasticity, Proc. Natl. Acad. Sci. 116 (52) (2019) 26414–26420.
[39] C. Settgast, M. Abendroth, M. Kuna, Constitutive modeling of plastic deformation behavior of open-cell foam structures using neural networks, Mech. Mater. 131 (2019) 1–10.
[40] Q. Rong, H. Wei, X. Huang, H. Bao, Predicting the effective thermal conductivity of composites from cross sections images using deep learning methods, Compos. Sci. Technol. 184 (2019) 107861.
[41] B. Mortazavi, I.S. Novikov, E.V. Podryabinkin, S. Roche, T. Rabczuk, A.V. Shapeev, X. Zhuang, Exploring phononic properties of two-dimensional materials using machine learning interatomic potentials, Appl. Mater. Today 20 (2020) 100685.
[42] K.M. Hamdia, H. Ghasemi, Y. Bazi, H. AlHichri, N. Alajlan, T. Rabczuk, A novel deep learning based method for the computational material design of flexoelectric nanostructures with topology optimization, Finite Elem. Anal. Des. 165 (2019) 21–30.
[43] S. Feng, H. Zhou, H. Dong, Using deep neural network with small dataset to predict material defects, Mater. Des. 162 (2019) 300–310.
[44] A.D. Spear, S.R. Kalidindi, B. Meredig, A. Kontsos, J.-B. Le Graverend, Data-driven materials investigations: the next frontier in understanding and predicting fatigue behavior, JOM 70 (7) (2018) 1143–1146.
[45] E. Ulu, R. Zhang, L.B. Kara, A data-driven investigation and estimation of optimal topologies under variable loading configurations, Comput. Methods Biomech. Biomed. Eng. 4 (2) (2016) 61–72.
[46] C. Yang, Y. Kim, S. Ryu, G.X. Gu, Prediction of composite microstructure stress-strain curves using convolutional neural networks, Mater. Des. 189 (2020) 108509.
[47] T. Nguyen, A. Kashani, T. Ngo, S. Bordas, Deep neural network with high-order neuron for the prediction of foamed concrete strength, Comput. Aid. Civil Infrastruct. Eng. 34 (4) (2019) 316–332.
[48] C.-T. Chen, G.X. Gu, Generative deep neural networks for inverse materials design using backpropagation and active learning, Adv. Sci. 7 (5) (2020) 1902607.
[49] V.M. Nguyen-Thanh, X. Zhuang, T. Rabczuk, A deep energy method for finite deformation hyperelasticity, Eur. J. Mech. A 80 (2020) 103874.
[50] M. Raissi, P. Perdikaris, G.E. Karniadakis, Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, J. Comput. Phys. 378 (2019) 686–707.
[51] E. Samaniego, C. Anitescu, S. Goswami, V.M. Nguyen-Thanh, H. Guo, K. Hamdia, X. Zhuang, T. Rabczuk, An energy approach to the solution of partial differential equations in computational mechanics via machine learning: concepts, implementation and applications, Comput. Methods Appl. Mech. Eng. 362 (2020) 112790.
[52] M.J. Jakiela, C. Chapman, J. Duda, A. Adewuya, K. Saitou, Continuum structural topology design with genetic algorithms, Comput. Methods Appl. Mech. Eng. 186 (2–4) (2000) 339–356.
[53] A. Kaveh, B. Hassani, S. Shojaee, S.M. Tavakkoli, Structural topology optimization using ant colony methodology, Eng. Struct. 30 (9) (2008) 2559–2565.
[54] Y. Maharaj, K.A. James, Metamaterial topology optimization of nonpneumatic tires with stress and buckling constraints, Int. J. Numer. Methods Eng. 121 (7) (2020) 1410–1439.
[55] D.W. Abueidda, S. Koric, N.A. Sobh, Topology optimization of 2d structures with nonlinearities using deep learning, Comput. Struct. 237 (2020) 106283.
[56] R. Alberdi, K. Khandelwal, Design of periodic elastoplastic energy dissipating microstructures, Struct. Multidiscip. Optim. 59 (2) (2019) 461–483.
[57] K.A. James, H. Waisman, Topology optimization of viscoelastic structures using a time-dependent adjoint method, Comput. Methods Appl. Mech. Eng. 285 (2015) 166–187.
[58] Y. Yu, T. Hur, J. Jung, I.G. Jang, Deep learning for determining a near-optimal topological design without any iteration, Struct. Multidiscip. Optim. 59 (3) (2019) 787–799.
[59] Q. Lin, J. Hong, Z. Liu, B. Li, J. Wang, Investigation into the topology optimization for conductive heat transfer based on deep learning approach, Int. Commun. Heat Mass Transf. 97 (2018) 103–109.
[60] S. Rawat, M.H. Shen, Application of Adversarial Networks for 3d Structural Topology Optimization, Tech. Rep, 2019 SAE Technical Paper.
[61] S. Rawat, M.-H. H. Shen, A novel topology optimization approach using conditional deep learning, arXiv (preprint arXiv:1901.04859).
[62] Y. Zhang, A. Chen, B. Peng, X. Zhou, D. Wang, A deep convolutional neural network for topology optimization with strong generalization ability, arXiv (preprint arXiv: 1901.07761).
[63] Y. Zhang, L. Gao, M. Xiao, Maximizing natural frequencies of inhomogeneous cellular structures by kriging-assisted multiscale topology optimization, Comput. Struct. 230 (2020) 106197.
[64] Y. Zhang, M. Xiao, X. Zhang, L. Gao, Topological design of sandwich structures with graded cellular cores by multiscale optimization, Comput. Methods Appl. Mech. Eng. 361 (2020) 112749.
[65] J. Guedes, N. Kikuchi, Preprocessing and postprocessing for materials based on the homogenization method with adaptive finite element methods, Comput. Methods Appl. Mech. Eng. 83 (2) (1990) 143–198.
[66] M. Ostoja-Starzewski, Microstructural Randomness and Scaling in Mechanics of Materials, CRC Press, 2007.
[67] Z. Xia, C. Zhou, Q. Yong, X. Wang, On selection of repeated unit cell model and application of unified periodic boundary conditions in micro-mechanical analysis of composites, Int. J. Solids Struct. 43 (2) (2006) 266–278.
[68] M.P. Bendsøe, Optimal shape design as a material distribution problem, Struct. Optim. 1 (4) (1989) 193–202.
[69] O. Sigmund, Morphology-based black and white filters for topology optimization, Struct. Multidiscip. Optim. 33 (4–5) (2007) 401–424.
[70] O. Sigmund, J. Petersson, Numerical instabilities in topology optimization: a survey on procedures dealing with checkerboards, mesh-dependencies and local minima, Struct. Optim. 16 (1) (1998) 68–75.
[71] C.S. Jog, R.B. Haber, Stability of finite element models for distributed-parameter optimization and topology design, Comput. Methods Appl. Mech. Eng. 130 (3–4) (1996) 203–226.
[72] T.E. Bruns, D.A. Tortorelli, Topology optimization of non-linear elastic structures and compliant mechanisms, Comput. Methods Appl. Mech. Eng. 190 (26–27) (2001) 3443–3459.
[73] L. Xia, P. Breitkopf, Design of materials using topology optimization and energy-based homogenization approach in matlab, Struct. Multidisc. Osptim. 52 (6) (2015) 1229–1241.
[74] Y. Wang, Z. Luo, N. Zhang, Z. Kang, Topological shape optimization of microstructural metamaterials using a level set method, Comput. Mater. Sci. 87 (2014) 178–186.
[75] L. Yin, W. Yang, Optimality criteria method for topology optimization under multiple constraints, Comput. Struct. 79 (20—21) (2001) 1839–1850.
[76] E. Andreassen, B.S. Lazarov, O. Sigmund, Design of manufacturable 3d extremal elastic microstructure, Mech. Mater. 69 (1) (2014) 1–10.
[77] Z. Zhang, Q. Liu, Y. Wang, Road extraction by deep residual u-net, IEEE Geosci. Remote Sens. Lett. 15 (5) (2018) 749–753.

[78] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, International Conference on Medical image computing and computer-assisted intervention, Springer 2015, pp. 234–241.
[79] F. Chollet, et al., Keras, https://github.com/fchollet/keras 2015.
[80] M. Abadi, A. Agarwal, P. Barham, E. Brevdo, Z. Chen, C. Citro, G.S. Corrado, A. Davis, J. Dean, M. Devin, S. Ghemawat, I. Goodfellow, A. Harp, G. Irving, M. Isard, Y. Jia, R. Jozefowicz, L. Kaiser, M. Kudlur, J. Levenberg, D. Mané, R. Monga, S. Moore, D. Murray, C. Olah, M. Schuster, J. Shlens, B. Steiner, I. Sutskever, K. Talwar, P. Tucker, V. Vanhoucke, V. Vasudevan, F. Viégas, O. Vinyals, P. Warden, M. Wattenberg, M. Wicke, Y. Yu, X. Zheng, TensorFlow: Large-scale Machine Learning on Heterogeneous Systems, software available from tensorflow.org 2015.
[81] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, arXiv (preprint arXiv:1412.6980).
[82] G.E. Hinton, A practical guide to training restricted boltzmann machines, Neural Networks: Tricks of the Trade, Springer 2012, pp. 599–619.
[83] K.H. Zou, S.K. Warfield, A. Bharatha, C.M. Tempany, M.R. Kaus, S.J. Haker, W.M. Wells III, F.A. Jolesz, R. Kikinis, Statistical validation of image segmentation quality based on a spatial overlap index1: scientific reports, Acad. Radiol. 11 (2) (2004) 178–189.
[84] S. Amstutz, S. Giusti, A. Novotny, E. de Souza Neto, Topological derivative for multi-scale linear elasticity models applied to the synthesis of microstructures, Int. J. Numer. Methods Eng. 84 (6) (2010) 733–756.
[85] O. Sigmund, A new class of extremal composites, J. Mech. Phys. Solids 48 (2) (2000) 397–428.
[86] Y. Zhang, M. Xiao, H. Li, L. Gao, Topology optimization of material microstructures using energy-based homogenization method under specified initial material layout, J. Mech. Sci. Technol. 33 (2) (2019) 677–693.
[87] O. Sigmund, A 99 line topology optimization code written in matlab, Struct. Multidiscip. Optim. 21 (2) (2001) 120–127.
[88] V.-N. Hoang, P. Tran, V.-T. Vu, H. Nguyen-Xuan, Design of lattice structures with direct multiscale topology optimization, Compos. Struct. 252 (2020) 112718.
[89] P. Liu, Z. Kang, Y. Luo, Two-scale concurrent topology optimization of lattice structures with connectable microstructures, Addit. Manufact. 36 (2020) 101427.
[90] S. Koric, A. Gupta, Sparse matrix factorization in the implicit finite element method on petascale architecture, Comput. Methods Appl. Mech. Eng. 302 (2016) 281–292.
[91] R. Borrell, J.C. Cajas, D. Mira, A. Taha, S. Koric, M. Vázquez, G. Houzeaux, Parallel mesh partitioning based on space filling curves, Comput. Fluids 173 (2018) 264–272.